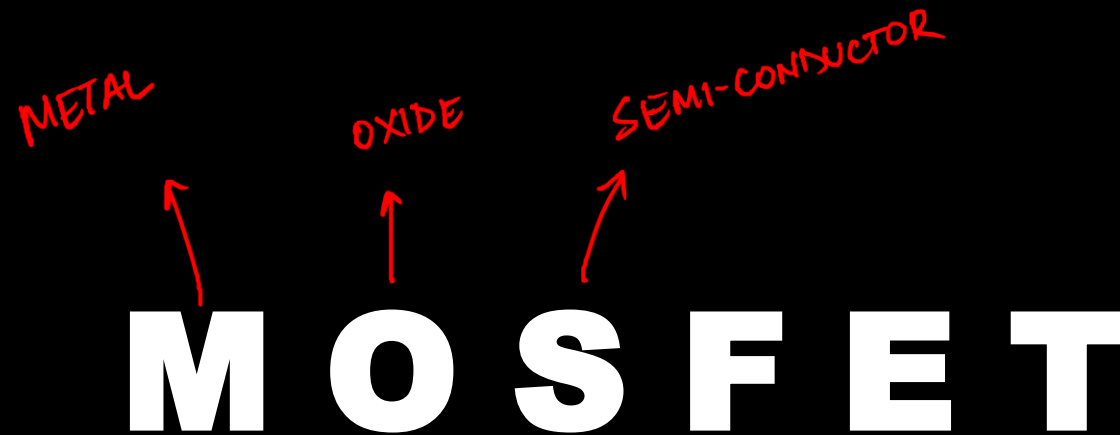




A diagram showing the acronym MOSFET in large white letters. Above the letters, three red handwritten labels with arrows point to specific parts: 'METAL' points to the 'M', 'OXIDE' points to the 'O', and 'SEMI-CONDUCTOR' points to the 'S'.

# M O S F E T

LECTURE – 1 :: (24-03-2025)

Er. Sailesh Mishra

[ Spec. Microelectronics & VLSI ]

Silicon University, Bhubaneswar

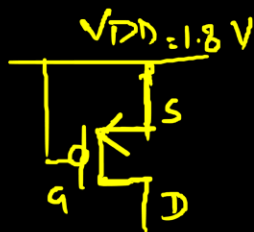
# MOSFET

## ANALOG

① As a Constant Current Source  $\left[ \uparrow \right] \rightarrow \left[ \frac{D}{G} \right]_s \rightarrow \text{sat}^n$

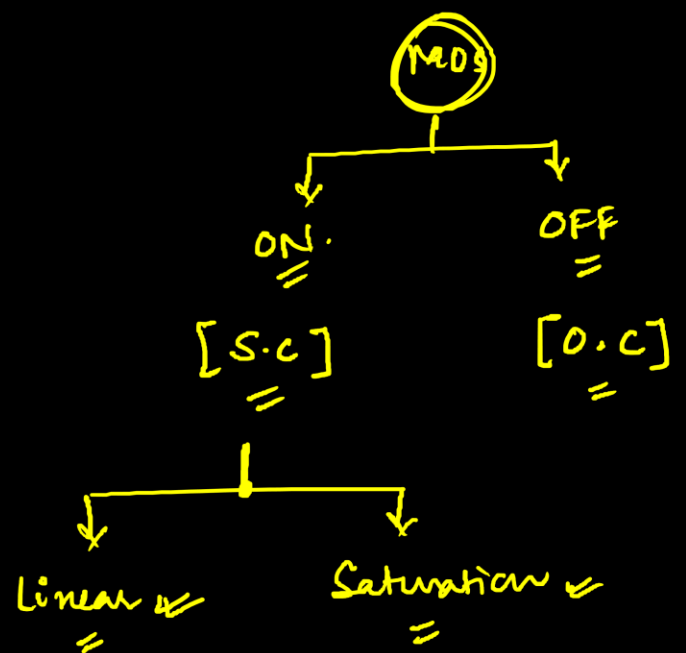
② As a Voltage-controlled Resistor (MOS  $\rightarrow$  LINEAR)  
or  
TRIODE region.

③ As a DIODE-CONNECTED  
LOAD

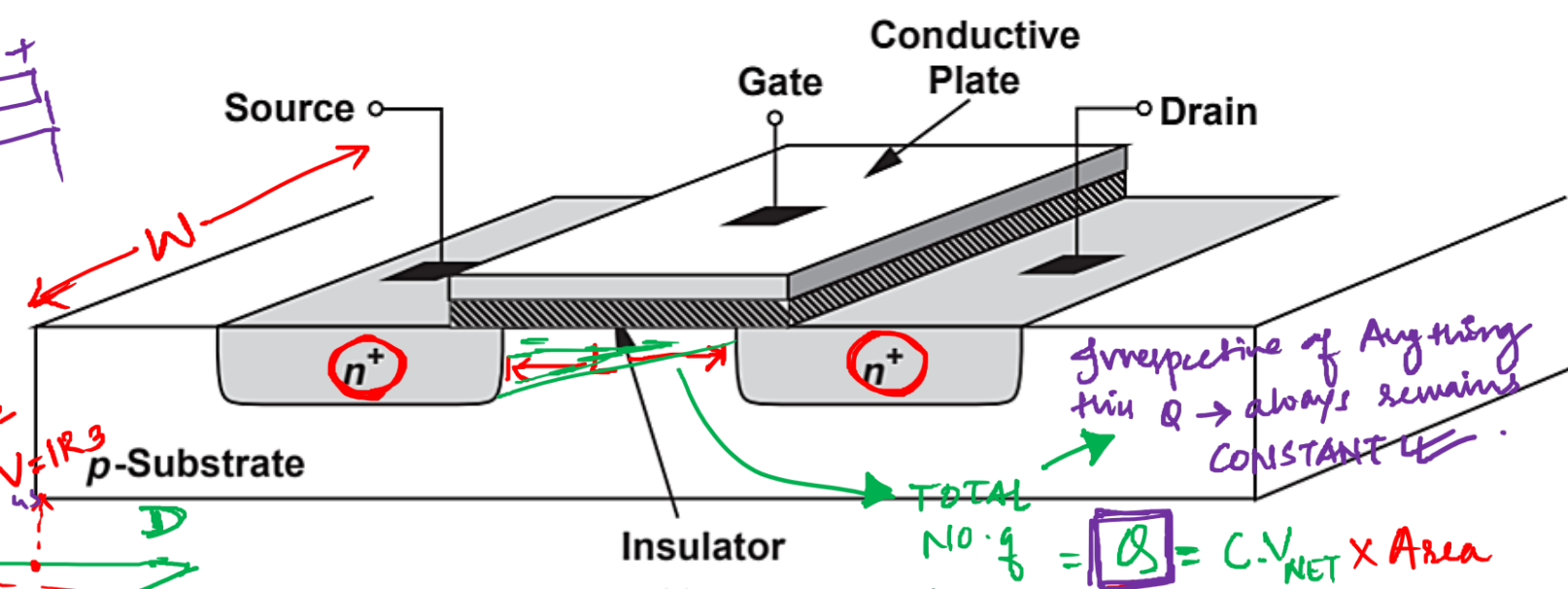
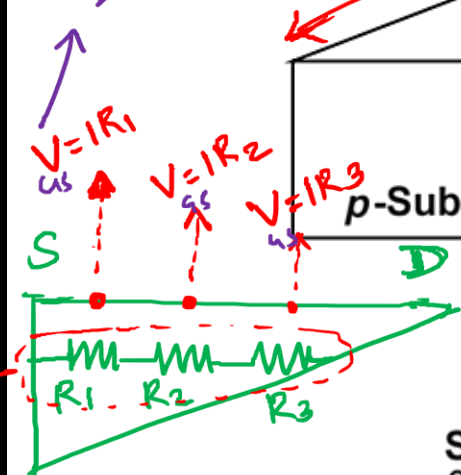
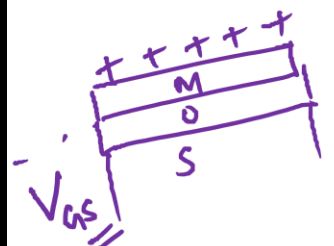


## DIGITAL

① As a SWITCH  $\rightarrow$  ON/OFF



"SERIES" =



Irrespective of Anything  
thin  $Q \rightarrow$  always remains  
CONSTANT  $\epsilon$ .

TOTAL  
No. of  
charges =  $Q = C \cdot V_{NET} \times \text{Area}$   
 $= C \cdot (V_{GS} - V_t) \times A$   
 $= C \cdot (V_{OV}) \times (W \times L)$

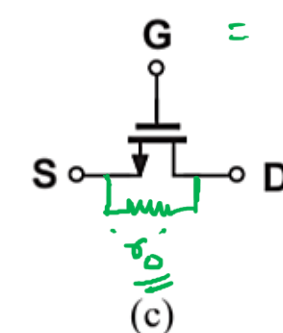
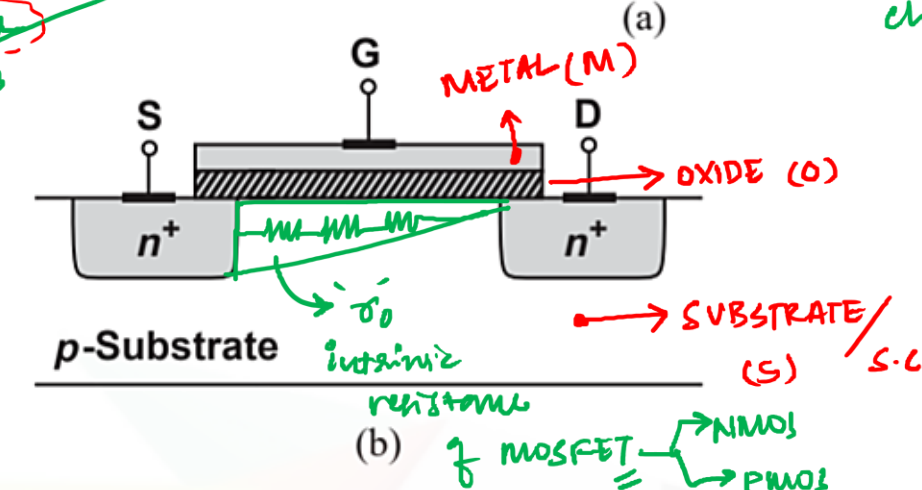
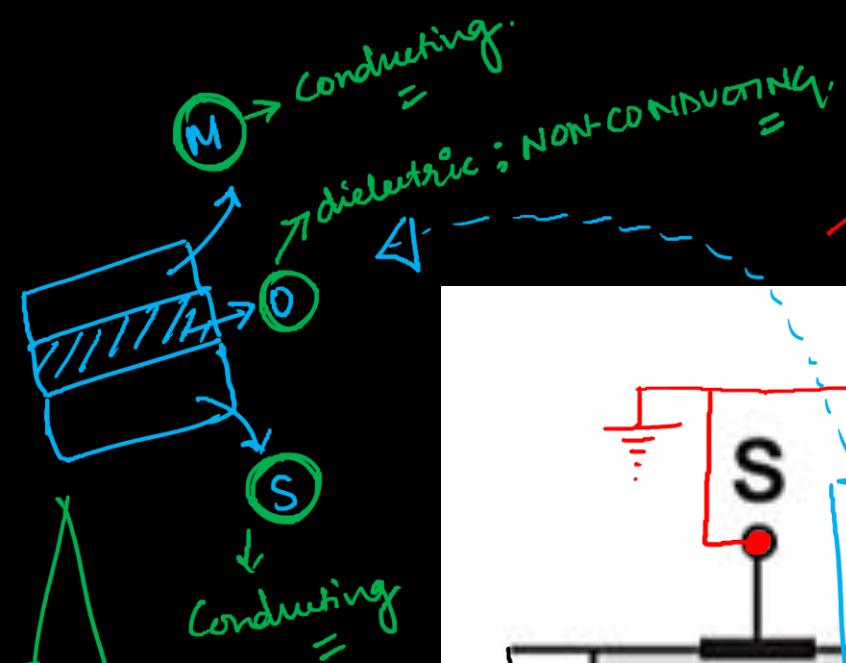
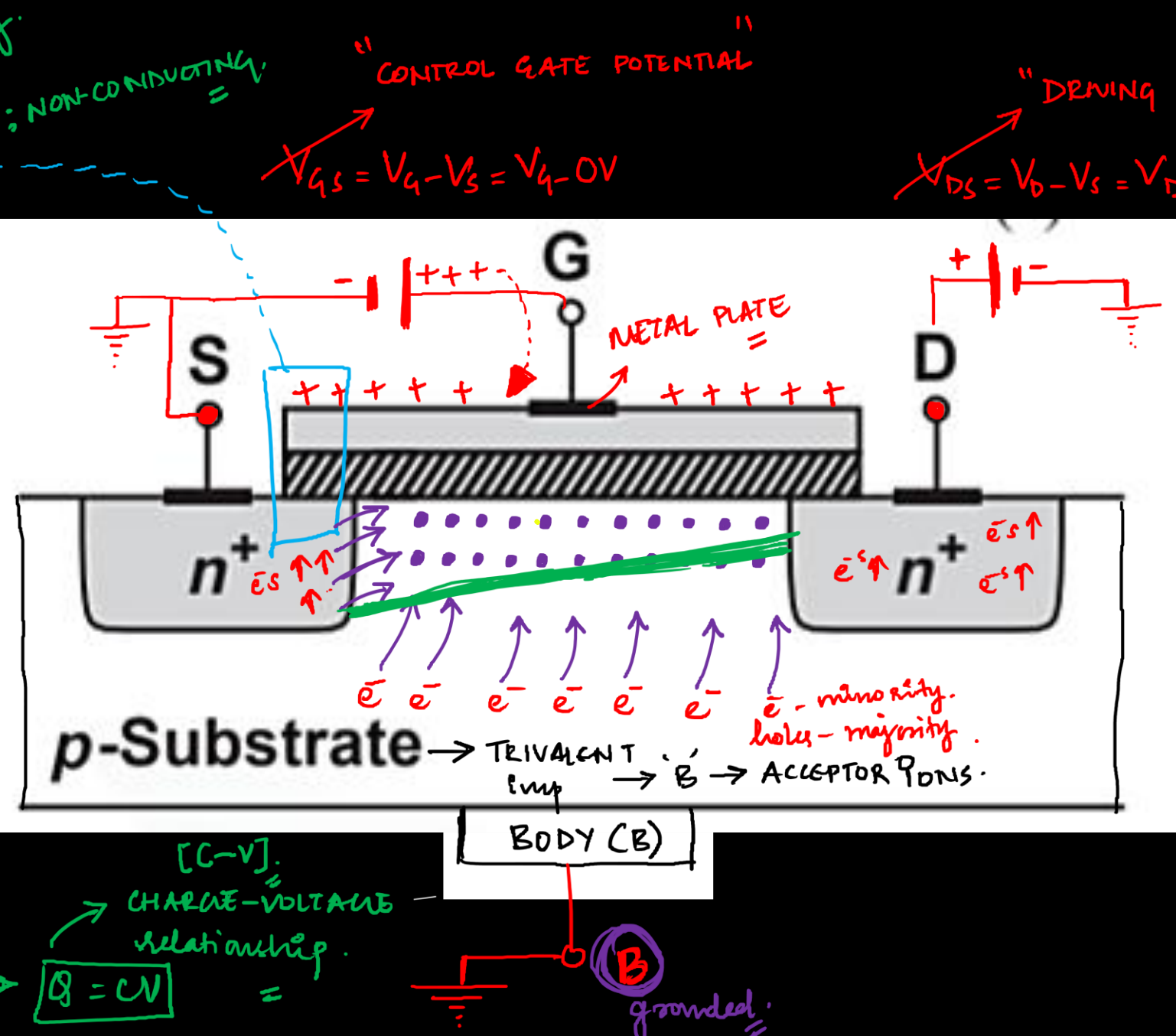
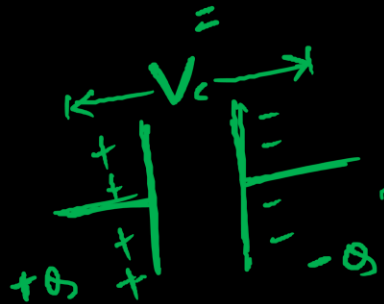


Figure 6.2 (a) Structure of MOSFET, (b) side view, (c) circuit symbol.

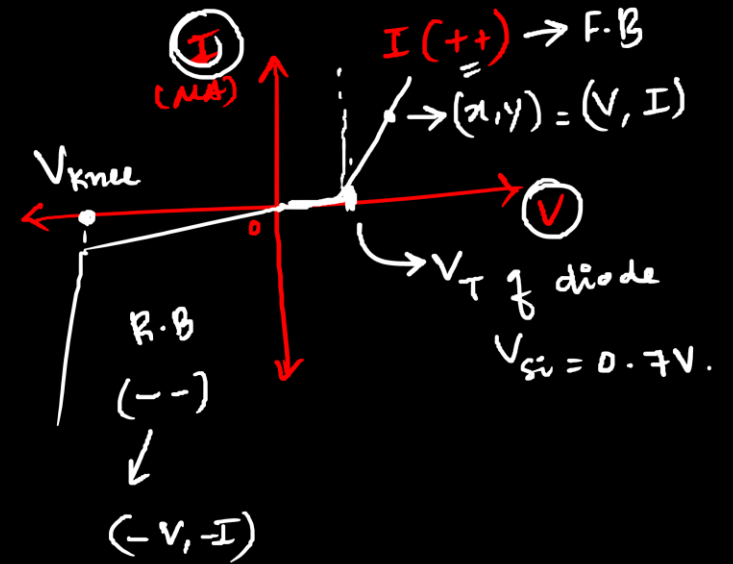
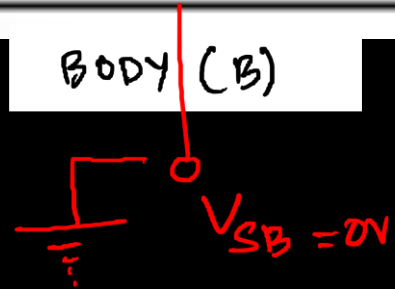
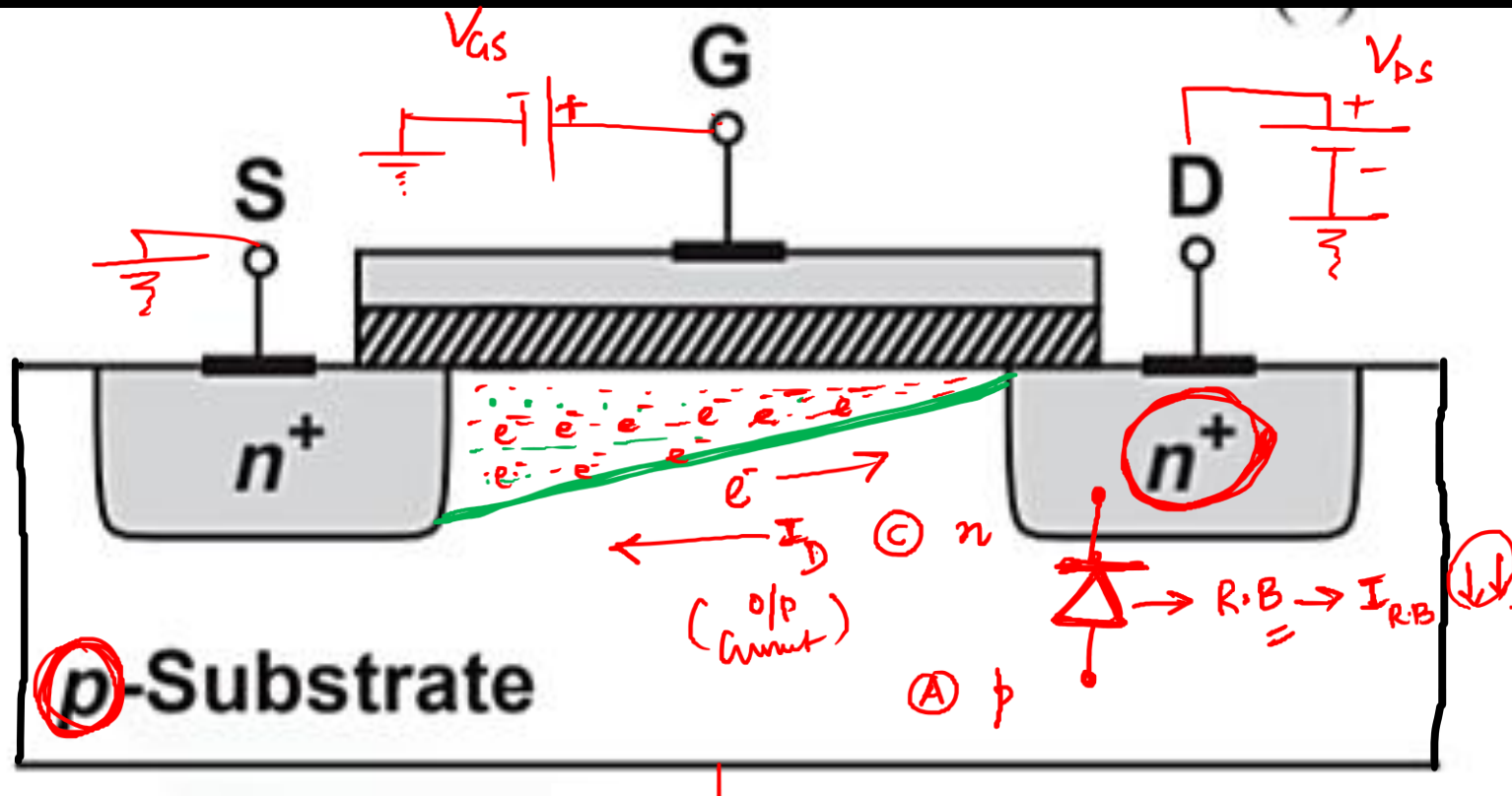
→ CHANNEL-  
ENHANCEMENT NMOS =



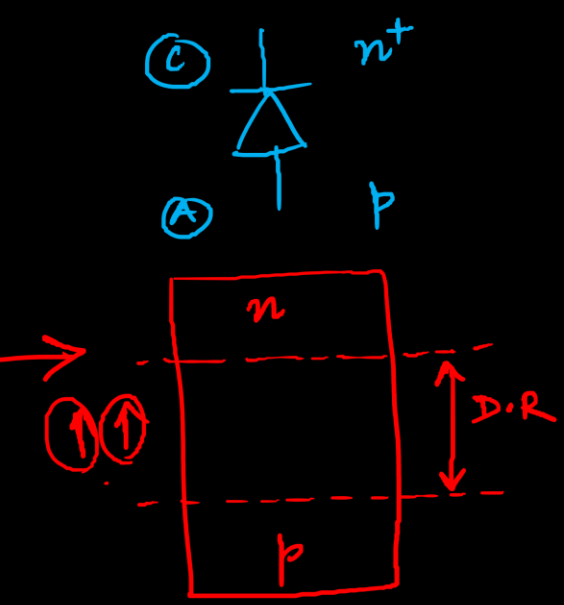
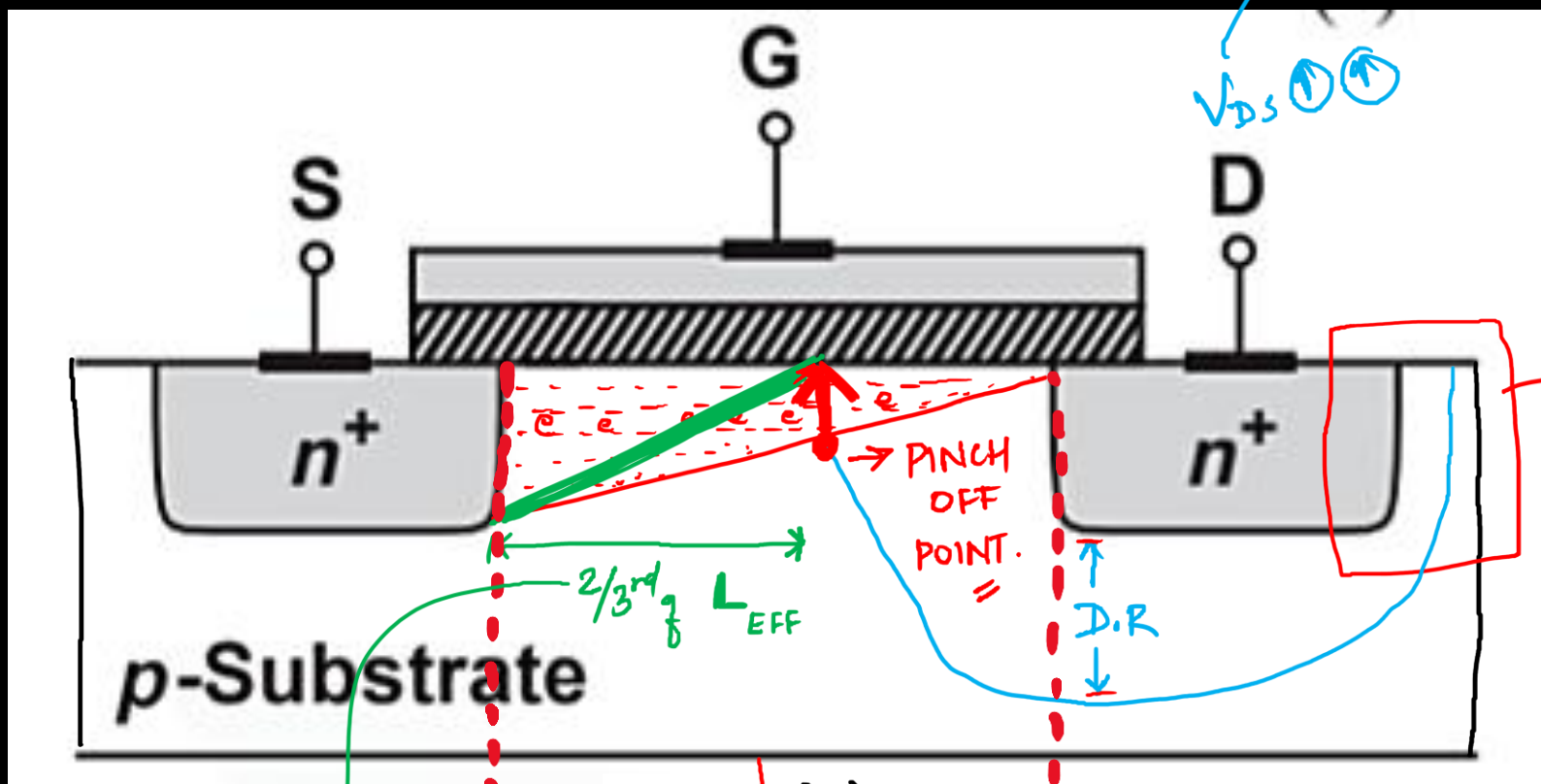
PARALLEL PLATE CAPACITOR.



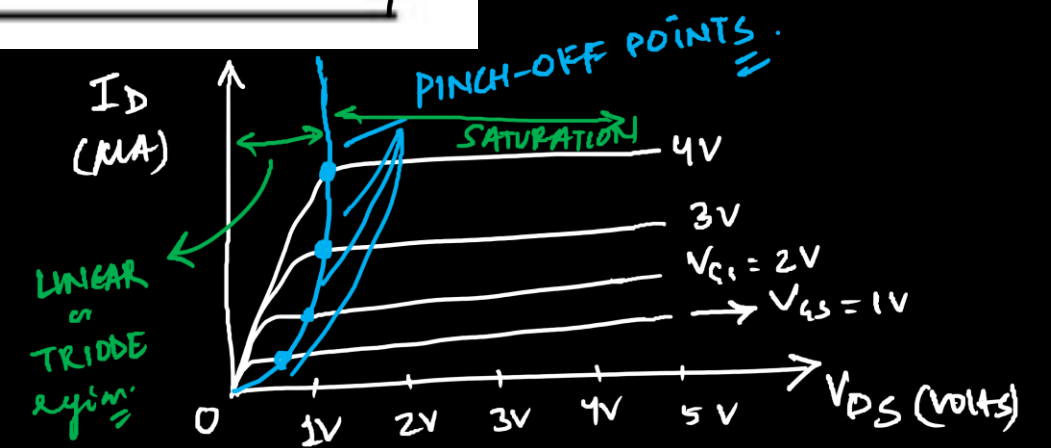
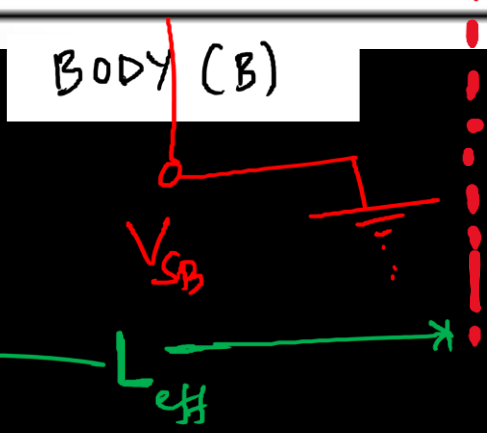
①  $V_{GS} = +1V$   
 $V_{DS} = 0V$  (const).  
 $V_{GS} \rightarrow 2V \rightarrow 3V \rightarrow 4V$   
 ②  $V_{DS} \rightarrow 0V \rightarrow \text{const}$   
 → CHARGES will get Accumulated just under the Gate Electrode  $G'$ .



; Magnitude will be very very less. R-C  
 ① F.B  $\rightarrow$  d  $\downarrow$   $\rightarrow$  ① C =  $\epsilon_0 \cdot A \cdot \frac{d}{d}$   $\rightarrow$  delay  $\uparrow$   
 ②  $I \neq 0$  MA (Reverse Situation Forward Current will flow into the device and hence DEVICE WILL DIS-INTEGRATE.

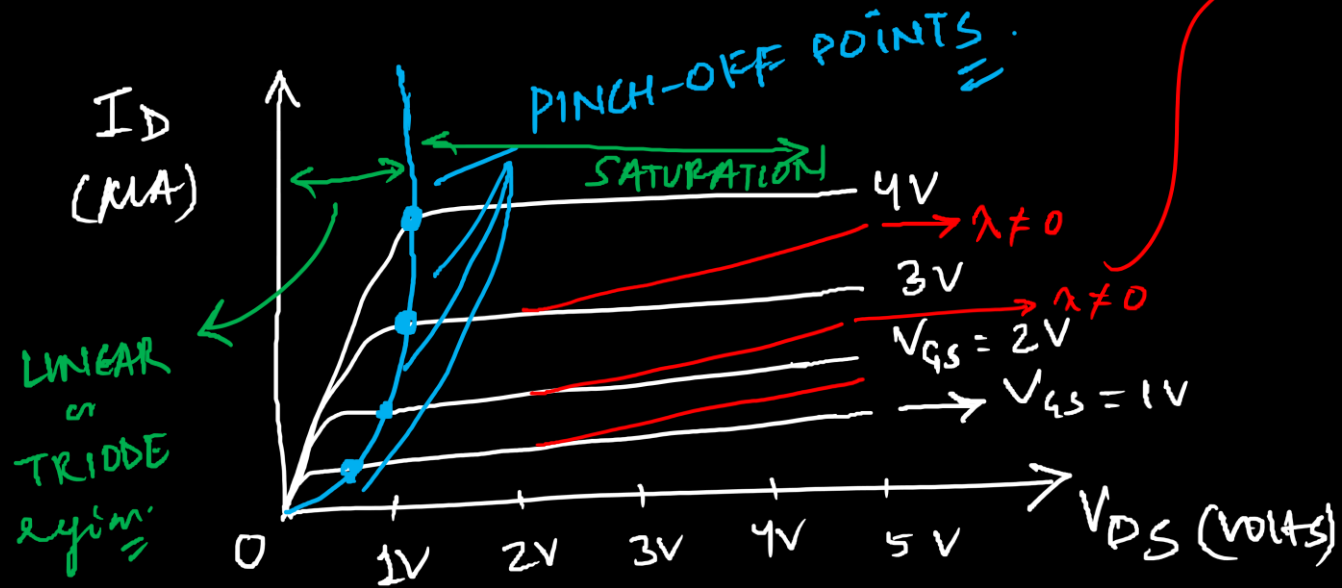


$Q_s = C \cdot V_n \times \text{Area}$   
 $= C \times V_{ov} \times W \times L_{eff}$



$I_D$  in MOSFET

- CUT-OFF. = 0mA
- LINEAR =  $\neq$  0mA
- SATURATION  $\neq$  0mA (1)



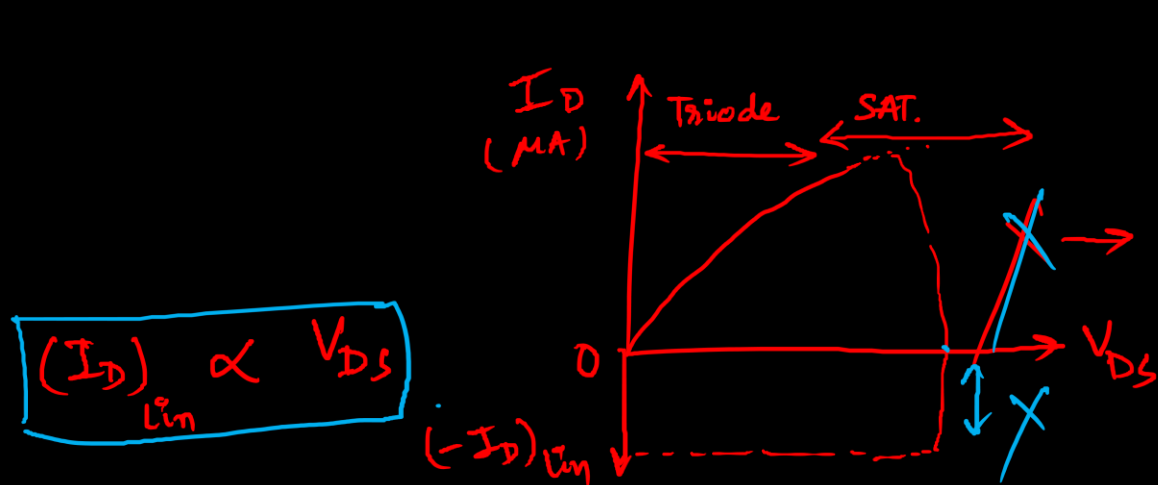
$(I_D)_{linear} = \underbrace{\mu_n C_{ox}}_{\text{TRANSISTOR ASPECT RATIO}} \cdot \underbrace{\frac{W}{L}}_{\text{OVERDRIVE VOLTAGE}} \left\{ (V_{GS} - V_{th}) V_{DS} - \frac{V_{DS}^2}{2} \right\}$

$K_n: \text{NMOS} [\mu A/V^2]$

$(I_D)_{sat} = \mu_n C_{ox} \cdot \frac{W}{L} \left\{ (V_{GS} - V_{th})^2 \right\}$  (1)  $\lambda = 0$

$\mu_n C_{ox}$

Square law



(1)  $(I_D)_{sat} = \mu_n C_{ox} \cdot \frac{W}{L} \left\{ (V_{GS} - V_{th})^2 (1 + \lambda V_{DS}) \right\}$  (1)

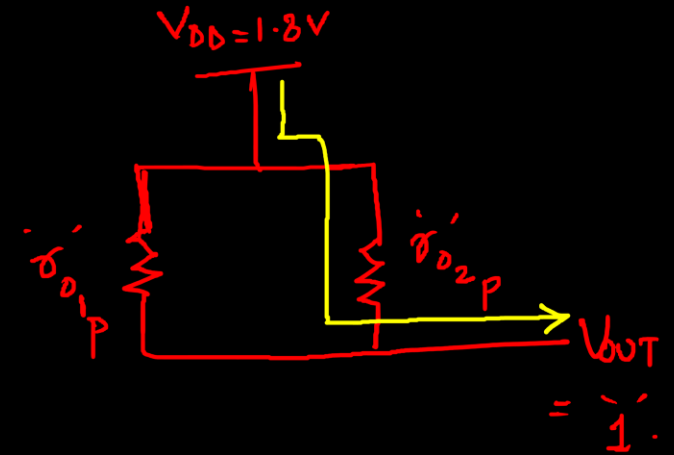
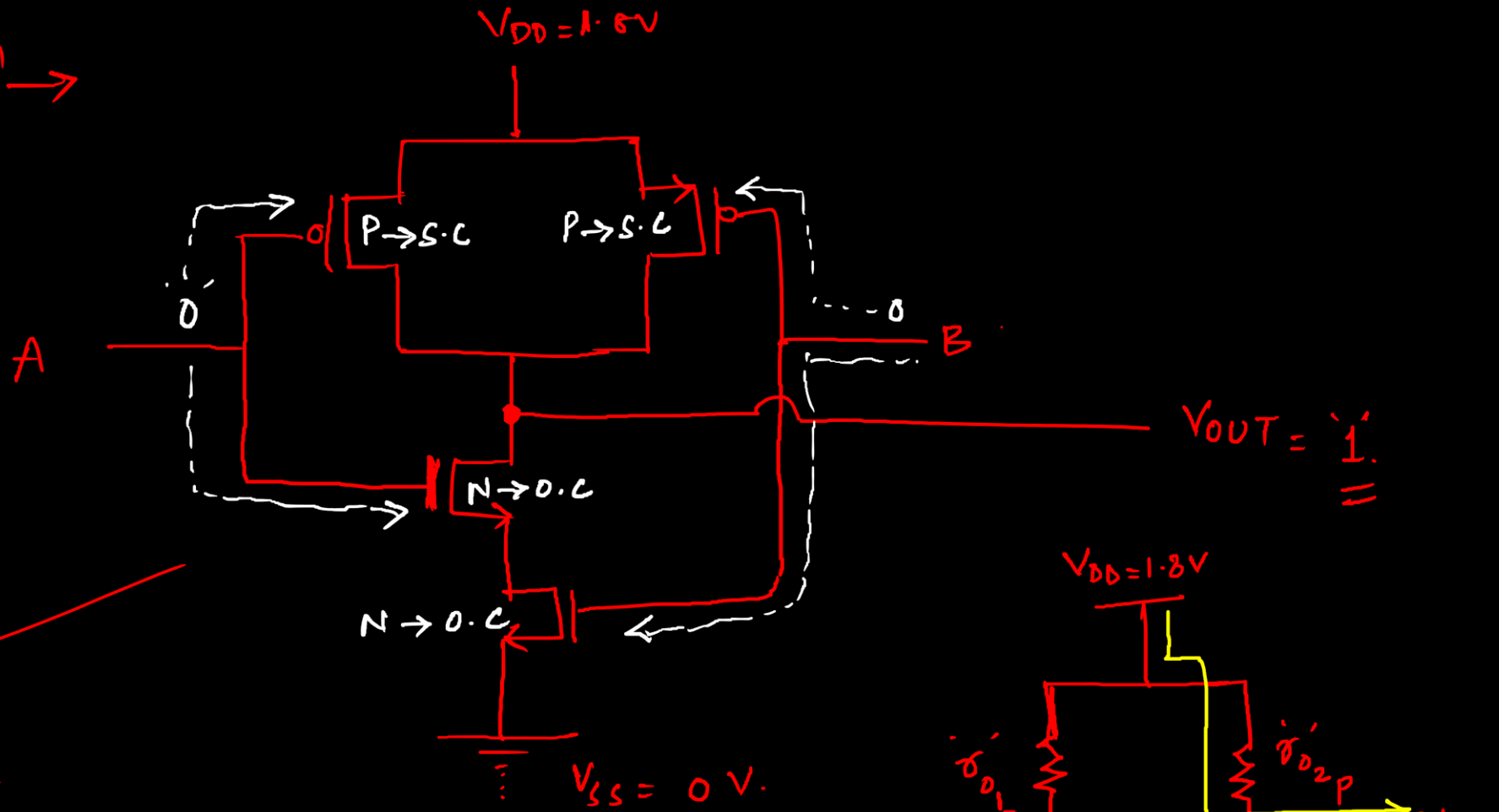
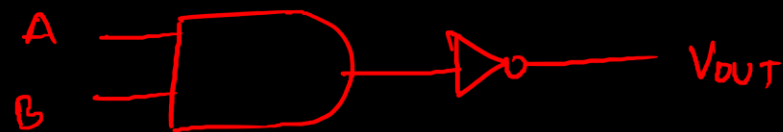
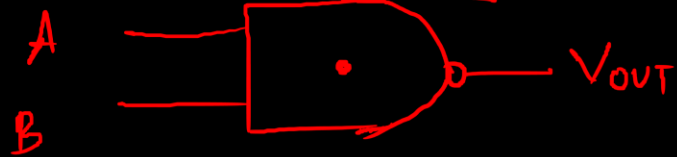
$\lambda \neq 0$

NOTE. # =

If  $V_{DS} \rightarrow \infty$ ;  $(I_D)_{lin} = 0 - \infty = \boxed{-ve} \dots$

NAND gate  $\rightarrow$  CMOS realiz<sup>n</sup>  $\rightarrow$

| A | B | $V_{out}$ |
|---|---|-----------|
| 0 | 0 | 1         |
| 0 | 1 | 1         |
| 1 | 0 | 1         |
| 1 | 1 | 0         |



NAND gate  $\rightarrow$  CMOS realiz<sup>n</sup>  $\rightarrow$

| A | B | $V_{out}$ |
|---|---|-----------|
| 0 | 0 | 1         |
| 0 | 1 | 1         |
| 1 | 0 | 1         |
| 1 | 1 | 0         |

